Name: Quiz 3 — 7/23

1. Find the general solution to the equation $y' = x^2/y^3$.

2. Consider the sequence $(a_n) = \left(\frac{2}{1}, \frac{3}{4}, \frac{4}{9}, \frac{5}{16}, \frac{6}{25}, \ldots\right)$. Find a formula for the nth term a_n. Compute $\lim_{n\to\infty} a_n$.

3. Compute $\displaystyle\sum_{n=1}^{\infty} \frac{5^{n-1}}{2^n}$.

4. Does the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+4}$ converge or diverge?